MIRROR: A Hierarchical Benchmark for Metacognitive Calibration in Large Language Models

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Abstract

We introduce MIRROR (Metacognitive Integrity and Recursive Reasoning in Operationalized Response), a benchmark comprising eight experiments across four metacognitive levels that evaluates whether large language models can use self-knowledge to make better decisions. We evaluate 16 models from 8 labs across approximately 250,000 evaluation instances using five independent behavioral measurement channels. We find two phenomena with direct implications for agentic deployment: (1) compositional self-prediction fails universally—the Metacognitive Convergence Index is exactly 0.000 across all 15 models tested, and (2) models possess accurate domain-specific self-knowledge but systematically fail to translate it into appropriate agentic action-selection—external metacognitive control reduces the Confident Failure Rate from 0.562 to 0.214, a 62% reduction. Providing models with their own calibration scores produces no significant improvement ($p > 0.05$); only architectural constraint is effective. The Knowing-Doing Index is negative for 9 of 10 models in the capstone experiment. These findings challenge the assumption underlying current agentic architectures that accurate self-knowledge enables self-correction, and suggest that external metacognitive scaffolding—not improved self-knowledge—is the path to safer autonomous AI systems. MIRROR and all data are publicly available.

1 Introduction

The deployment of large language models (LLMs) as autonomous agents presupposes a form of metacognitive competence: the model must not only know *what* it knows, but must translate that awareness into appropriate action—proceeding when confident, escalating to tools or human review when uncertain. This assumption underlies agentic architectures built on self-reflection loops, confidence-based routing, and Constitutional AI self-monitoring [Perez et al., 2022]. It is foundational to the enterprise of agentic AI safety. It has never been systematically tested.

Prior work has studied LLM calibration [Kadavath et al., 2022, Kapoor et al., 2024], uncertainty estimation [Lin et al., 2024], and situational awareness [Laine et al., 2024]. But these address different questions: calibration measures whether confidence matches accuracy; situational awareness measures whether models know *what they are*. Nobody has systematically tested whether models can *use* their self-knowledge to make better decisions—whether the bridge from metacognitive *monitoring* to metacognitive *control* [Nelson and Narens, 1990] is intact.

We introduce MIRROR, a hierarchical benchmark spanning four metacognitive levels (atomic self-knowledge, cross-domain transfer, compositional prediction, and adaptive self-regulation), eight experiments, and five independent behavioral measurement channels. We evaluate 16 models from 8 labs across approximately 250,000 evaluation instances.[1]

Three findings emerge. First, compositional self-prediction fails universally: the Metacognitive Convergence Index (MCI) is exactly 0.000 across all 15 models tested—models that calibrate

[1]Computed as: Exp1 (5,000 $\times$ 5 channels $\times$ 16 models) + Exp4 (320 $\times$ 2 conditions $\times$ 15 models) + Exp9 (597 $\times$ 4 conditions $\times$ 3 paradigms $\times$ 10 models) + Experiments 2, 3, 5, 6, 8 ($\sim$30,000 total). See Table 3 for per-experiment counts.

Table 1: Comparison of MIRROR with prior benchmarks for LLM self-knowledge and metacognition.

Feature	SAD (Laine 2024)	Barkan et al. (2025)	Kadavath et al. (2022)	Steyvers et al. (2025)	MIRROR (Ours)
Venue	NeurIPS D&B '24	NeurIPS WS '25	arXiv	arXiv	NeurIPS D&B '26
Focus	Situational awareness	Self-capability prediction	Calibration (P(IK))	Metacognition + uncertainty	Metacognitive cal. → agentic action
Models	16	∼15	∼10	2 families	16
Task categories	7	3	—	6 datasets	8 experiments
Total items	13K questions	∼1K tasks	—	12K+ items	∼250K records
Metacognitive levels	1 (awareness)	1 (confidence)	1 (calibration)	1 (calibration)	4 (L0–L3)
Behavioral channels	1 (MC/free)	1 (verbal)	1 (verbal)	1 (verbal)	5 (wager, opt-out, tool, diff., natural)
Agentic connection	No	Yes (SWE-bench)	No	No	Yes (4 cond., 3 par.)
Headline finding	Partial sit. awareness	LLMs over-confident	*Mostly* calibrated	Task-specific metacognition	Knowing-doing gap (62% CFR reduction)
Public dataset	Yes	No	No	No	Yes

accurately in individual domains completely fail to compose those calibrations for multi-domain tasks. Second, the knowing-doing gap: models possess accurate self-knowledge but do not act on it. External constraint reduces the Confident Failure Rate (CFR) from 0.562 to 0.214 (62% reduction); providing models with their own calibration scores produces no significant improvement. Third, self-knowledge is domain-atomic: the Transfer Influence Index ranges from 0.019 to 0.175 across 11 models, indicating that calibration does not meaningfully cross domain boundaries.

Contributions.

- The MIRROR benchmark: 8 experiments across 4 metacognitive levels, 5 behavioral channels, 5,000 domain-stratified questions, and 597 composite agentic tasks, publicly released with all protocols and analysis code.
- Two empirical findings—universal compositional failure (MCI = 0.000) and the knowing-doing gap (62% CFR reduction under external control)—that challenge assumptions about LLM self-knowledge.
- Quantification of the knowing-doing gap's cost: in a 1,000-task agentic deployment, ∼648 tasks fail silently without external constraint versus ∼214 with constraint.
- Evidence that external metacognitive scaffolding, not better self-knowledge, is the path to safer agentic systems.

Table 1 positions MIRROR relative to prior work.

2 Related Work

LLM Calibration. Kadavath et al. [2022] showed that LLMs can predict their own answer correctness above chance on knowledge-intensive tasks. Kapoor et al. [2024] demonstrated that fine-tuning is needed for reliable calibration. Steyvers et al. [2025] found that metacognitive ability is task-specific, not general—a finding consistent with MIRROR's domain-level analysis. MIRROR differs from all prior calibration work in testing whether calibration *transfers to action-selection*: we measure the gap between knowing and doing, not the accuracy of knowing alone.

LLM Self-Knowledge and Situational Awareness. SAD [Laine et al., 2024] measures whether LLMs know what they are—identity, lifecycle stage, deployment context. Barkan et al. [2025] tested capability prediction on SWE-bench tasks, finding systematic overconfidence. Xiong et al. [2024] found that LLMs are often overconfident when expressing numerical probability estimates. MIRROR is complementary: SAD diagnoses awareness; MIRROR diagnoses the gap between awareness and behavior.

Agentic Evaluation. AgentBench [Liu et al., 2024], OSWorld [Xie et al., 2024], and GAIA [Mialon et al., 2023b] measure task success rate in agentic settings. These benchmarks evaluate execution

quality conditional on the decision to act. Concurrent work on agentic confidence calibration [Anonymous, 2026a] measures calibration during multi-step trajectories using trajectory-level features. MIRROR differs in measuring metacognitive calibration *before* deployment and testing whether pre-deployment assessment predicts agentic outcomes. More broadly, MIRROR inverts the standard framing: we measure the *decision* itself, treating task execution quality as a downstream variable and calibration-informed decision-making as the primary outcome.

Metacognition in AI. Metacognitive prompting [Wang and Zhao, 2024] uses metacognitive strategies to improve LLM performance. Augmented language model surveys [Mialon et al., 2023a] document the tool-use landscape but do not address when tool use *should* occur. Concurrent work on metacognition in large reasoning models [Anonymous, 2026b] tests difficulty awareness, confidence adjustment, and strategy revision across reasoning models. MIRROR extends this by connecting metacognitive assessment to agentic deployment through the Experiment 9 escalation design, providing the first benchmark-scale empirical test linking metacognitive assessment to agentic deployment outcomes.

3 The MIRROR Benchmark

MIRROR decomposes LLM metacognition into four levels of increasing cognitive complexity, inspired by the hierarchical structure of human metacognitive monitoring and control [Flavell, 1979, Nelson and Narens, 1990]. At each level, we measure behavior through five independent channels to prevent format-specific gaming.

3.1 Metacognitive Levels

Level 0—Atomic Self-Knowledge (Calibration). Can the model accurately estimate $P(\text{correct})$ for a given question in a given domain? This is the best-studied level [Kadavath et al., 2022, Kapoor et al., 2024] and serves as MIRROR's foundation. Experiment 1 establishes per-model, per-domain calibration profiles across 8 cognitive domains.

Level 1—Cross-Domain Transfer. Does calibration in domain A influence behavior on tasks that implicitly require domain A skills? A model that knows it is weak at spatial reasoning (Level 0) should show increased caution on a trip-planning task that requires spatial reasoning, even though the surface framing is "travel advice." Experiments 2 and 3 test this transfer.

Level 2—Compositional Prediction. Can the model predict its performance on tasks requiring *combinations* of skills? A model that is 80% accurate on arithmetic and 70% on coding should estimate its accuracy on a task requiring both—but if self-knowledge is atomic rather than compositional, this prediction will fail. Experiment 3 tests this directly.

Level 3—Adaptive Self-Regulation. Does the model translate self-knowledge into appropriate action? This is the critical gap between metacognitive *monitoring* (knowing your limits) and metacognitive *control* (acting on that knowledge)—a distinction well-established in human cognition [Metcalfe and Finn, 2008]. Experiments 4, 5, 6, and 9 test whether self-knowledge drives behavior change, feedback adaptation, ecosystem interactions, and agentic decision-making.

We hypothesize these levels form an empirical difficulty ordering (L0 easiest, L3 hardest). Whether this ordering holds—and where it breaks—is itself a testable prediction.

3.2 Behavioral Measurement Channels

A single confidence elicitation format (*e.g.*, verbal "rate your confidence 1–10") is vulnerable to prompt-specific behavior. MIRROR measures metacognitive behavior through five independent channels, each requiring a separate API call with a fresh context:

1. **Wagering**—the model stakes points proportional to its confidence, with scoring that penalizes overconfident wrong answers.
2. **Opt-out**—the model may decline to answer. Skip rate should correlate with error rate.
3. **Difficulty selection**—given two questions (easy and hard), the model chooses which to attempt.

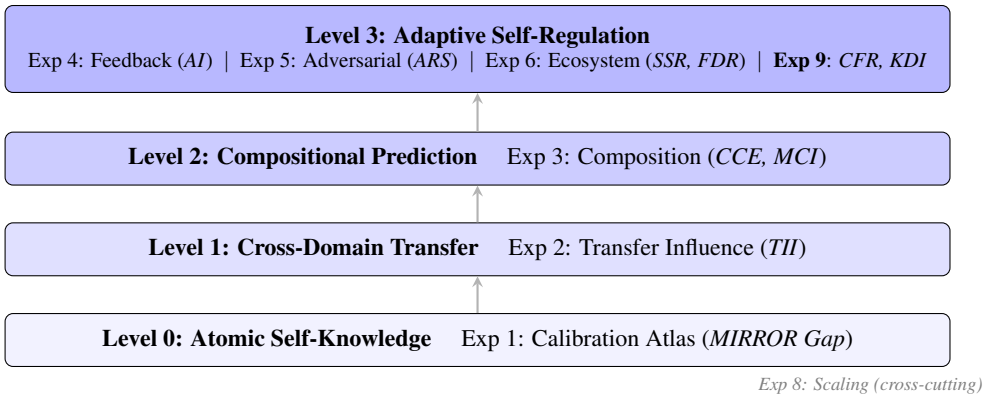

Figure 1: The MIRROR experimental hierarchy. Each level builds on measurements from the level below. Experiment 1 produces per-model calibration profiles consumed by all subsequent experiments. The capstone (Experiment 9) tests whether self-knowledge translates to appropriate agentic decisions.

4. **Tool delegation**—the model may invoke a calculator, search engine, or flag for human review rather than answering directly.
5. **Natural language signals**—hedging language, uncertainty markers, and response length, extracted from unconstrained free-form responses.

The **Metacognitive Convergence Index (MCI)** measures agreement across channels: do all five channels agree about which questions are hard for this model? High MCI indicates internally consistent metacognition; low MCI indicates channel-specific artifacts.

3.3 Experiments

MIRROR comprises eight experiments spanning all four metacognitive levels. Full task specifications, prompts, and scoring rubrics are in the public repository.

Experiment 1: Self-Knowledge Atlas (Level 0). 5,000 questions across 8 domains (arithmetic, spatial, temporal, linguistic, logical, social, factual, procedural), each measured through all 5 behavioral channels. Establishes per-model calibration profiles: which domains the model is well-calibrated in, and where its confidence diverges from accuracy. 16 models evaluated. Primary metrics: MIRROR gap (wagering accuracy − natural accuracy), MCI.

Experiment 2: Cross-Domain Transfer (Level 1). $\sim$150 tasks per model where the surface domain (*e.g.*, "travel planning") conceals a dependency on a hidden domain (*e.g.*, spatial reasoning). For each model, we compare behavioral confidence on the task to the model's Experiment 1 accuracy in the hidden domain. The Transfer Influence Index (TII) quantifies whether self-knowledge transfers across domain boundaries. 11 models evaluated. Primary metric: TII (range: 0.019–0.175, indicating weak-to-minimal transfer).

Experiment 3: Compositional Self-Prediction (Level 2). $\sim$25 tasks per model requiring combinations of two domains. Models predict their performance on the composite task, and we compare this prediction to their known accuracy in each component domain. The Compositional Calibration Error (CCE) measures prediction accuracy. 15 models evaluated. Primary finding: MCI = 0.000 across all models—compositional self-prediction is no better than chance.

Experiment 4: Adaptation Crucible (Level 3). 320 burn-and-test templates $\times$ 2 conditions (true feedback and sycophancy feedback) across 8 domains. The Adaptation Index (AI) measures whether feedback changes subsequent behavior. The Sycophancy Adaptation Ratio (SAR) measures

differential adaptation to false vs. true feedback. 15 models evaluated. Primary findings: AI ≈ 0 for most models (feedback immunity); SAR ranges from -1.52 to 2.97 (substantial sycophancy in a subset of models).

Experiment 5: Adversarial Robustness (Level 3). 320 adversarial trials + 320 matched clean controls per model. Four attack types (authority pressure, social proof, anchoring, framing) test whether metacognitive calibration degrades under adversarial manipulation. 14 models evaluated. Primary metric: ARS = 0.937–0.991 across all models—calibration is robust to framing attacks.

Experiment 6: Ecosystem Effect (Level 3). Three sub-experiments testing metacognition in social and evaluative contexts. (a) Multi-agent trust calibration: 115 scenarios $\times$ 4 conditions measuring sycophancy (SSR). (b) Flawed premise detection: 110 flawed + 110 well-formed tasks (FDR: 0.000–1.000, extreme inter-model variation). (c) Value robustness: 40 prompts per model testing resistance to incremental value erosion. 16 models evaluated.

Experiment 8: Scaling Analysis (Levels 0–3). Within-family comparison across Llama 3.1 8B $\rightarrow$ 70B $\rightarrow$ 405B and a generation comparison (Llama 3.1-70B vs. 3.3-70B). Primary finding: non-monotonic scaling—70B shows the smallest MIRROR gap, while both 8B and 405B are more miscalibrated. Calibration improves with generation (3.3 $>$ 3.1 at matched size) but not monotonically with parameters.

Experiment 9: The Knowing-Doing Gap (Level 3+). 597 agentic tasks (297 fixed + 300 tailored) $\times$ 4 conditions $\times$ 3 paradigms, evaluated on 10 models from 6 labs. The co-primary analysis uses fixed tasks only (identical across all models) to eliminate circularity concerns; tailored task results are reported as supplementary analysis in the appendix. Each task has two components requiring different domain skills. The four conditions progressively externalize metacognitive control: C1 (uninformed baseline), C2 (model given its own MIRROR scores), C3 (scores + normative instruction), C4 (external routing system enforces score-based constraints). Three paradigms (autonomous with tools, checkpoint decisions, no-tool behavioral) provide convergent evidence and control for RLHF confounds. Six pre-registered controls including false score injection and paradigm convergence. Primary metrics: CFR, KDI, Escalation Curve.

3.4 Evaluation Infrastructure

MIRROR is designed for reproducible, low-cost evaluation of new models. The benchmark requires only API access (no weights, no activations, no fine-tuning). Running the full benchmark on a new model requires approximately 8,000 API calls ($\sim$3 hours at 40 requests/minute). All prompts, scoring rubrics, and analysis code are publicly available. The dataset is hosted on HuggingFace with Croissant metadata.

4 Experimental Setup

We evaluate 16 models from 8 labs spanning 3B to 671B parameters (Table 2). All 16 models participate in the core experiments (1, 4, 5, 6); subsets participate in experiments with specific requirements: 11 models in Experiment 2 (requires cross-domain task infrastructure), 15 in Experiment 3, 10 in Experiment 9 (requires complete Experiment 1 profiles for tailored task generation), and 3 in Experiment 8 (Llama family scaling analysis). Two additional models were excluded: `command-r-plus` (API removed mid-collection) and `qwen-3-235b` (100% API failure rate on the NIM endpoint). All experiments use `temperature=0` for reproducibility. Infrastructure: NVIDIA NIM free tier (primary, for Llama, Mistral, Gemma, Phi-4, Kimi-K2), DeepSeek API (deepseek-r1/v3), and Google AI Studio (gemini-2.5-pro, gemma-3-12b). Concurrency: 32 parallel calls per model. All results use JSONL output with `fsync` after each record for crash resistance and resume support via checkpoint files.

Table 2: Model roster. All models are instruction-tuned variants accessed via API. Parameter counts are approximate where not officially disclosed.

Model	Lab	Params	Arch.	API	Experiments
`deepseek-r1`	DeepSeek	671B	Dense	DeepSeek	1–6, 9
`deepseek-v3`	DeepSeek	671B	Dense	DeepSeek	1–6
`gemini-2.5-pro`	Google	>1T	Dense	Google AI	1, 4, 6
`gemma-3-12b`	Google	12B	Dense	Google AI	1, 3–6
`gemma-3-27b`	Google	27B	Dense	NVIDIA NIM	1–6, 9
`gpt-oss-120b`	OpenAI	120B	Dense	NVIDIA NIM	1–6, 9
`kimi-k2`	Moonshot AI	200B	Dense	NVIDIA NIM	1–6, 9
`llama-3.1-405b`	Meta	405B	Dense	NVIDIA NIM	1–4, 6, 8–9
`llama-3.1-70b`	Meta	70B	Dense	NVIDIA NIM	1–6, 8–9
`llama-3.1-8b`	Meta	8B	Dense	NVIDIA NIM	1–6, 8–9
`llama-3.2-3b`	Meta	3B	Dense	NVIDIA NIM	1, 3–6
`llama-3.3-70b`	Meta	70B	Dense	NVIDIA NIM	1–6, 8–9
`mistral-large`	Mistral	675B	Dense	NVIDIA NIM	1–3, 5–6, 9
`mixtral-8x22b`	Mistral	193B	MoE (8x22B)	NVIDIA NIM	1, 3–6
`phi-4`	Microsoft	14B	Dense	NVIDIA NIM	1–6, 9
`qwen3-next-80b`	Alibaba	80B	Dense	NVIDIA NIM	1, 3–6

Table 3: Summary statistics for each MIRROR experiment.

Exp.	Description	Models	Items/Model	Primary Metric	M
Exp1	Atomic Self-Knowledge	16	330/model (8 domains x 5 subcats x 8 Qs)	MIRROR Gap	0
Exp2	Cross-Domain Transfer	11	150/model	TII	0
Exp3	Compositional Self-Prediction	15	25/model (mixed tasks)	CCE	0
Exp4	Adaptive Self-Regulation (Burn-a...	15	320/model x 2 conditions	AI	0
Exp5	Adversarial Robustness of Self-K...	14	320/model (adv) + 320/model (clean)	ARS	0
Exp6	Ecosystem Effect (Trust + Flawed...	16	240/model (6a+6b+6c)	FDR	0
Exp8	Scaling Analysis	4	80/model (gap-fill)	Scaling slope	0
Exp9	The Knowing-Doing Gap	10	597 tasks x 4 conditions x 3 paradigms	CFR	0

Table 3 summarizes the scale and primary metrics for each experiment.

5 Results

5.1 Main Results

Table 4 reports all primary metrics across 16 models and 8 experiments. Three patterns are immediately visible. First, the MIRROR gap is universally positive (mean = 0.170, range 0.074–0.267): all models are overconfident in the wagering channel relative to natural accuracy. Second, CFR in the uninformed condition ranges from 0.387 to 0.907—even the best model fails on 39% of weak-domain agentic tasks when proceeding autonomously. Third, KDI is negative for 9 of 10 models evaluated in Experiment 9, confirming that self-knowledge systematically fails to translate into appropriate action.

5.2 Finding 1: Compositional Self-Prediction Fails Universally

Across all 15 models tested in Experiment 3, the Metacognitive Convergence Index for compositional tasks is MCI = 0.000. Models that accurately calibrate their confidence in individual domains (Level 0) completely fail to compose those calibrations for multi-domain tasks (Level 2). This is not a gradual degradation—it is a categorical failure. A model that knows "I am 80% accurate on

Table 4: Main results across all experiments for 16 models. Nat.Acc = natural accuracy (Exp1), Wag.Acc = wagering accuracy (Exp1), M.Gap = MIRROR gap (Exp1), TII = Transfer Integrity Index (Exp2), CCE = Compositional Calibration Error for mixed tasks (Exp3), AI = Adaptation Index on wager channel (Exp4), SAR = Sycophancy Adaptation Ratio (Exp4), ARS = Adversarial Robustness Score (Exp5), FDR = Flaw Detection Rate (Exp6b), SSR = Sycophancy Separation Ratio (Exp6a), CFR_{C1} = Confident Failure Rate in uninformed condition (Exp9), KDI = Knowing-Doing Index (Exp9). "—" indicates the model was not evaluated in that experiment.

Model	Nat.Acc	Wag.Acc	M.Gap	TII	CCE	AI	SAR	ARS	FDR	SSR	CFR_{C1}	KDI
deepseek-r1	0.437	0.569	0.139	0.175	0.511	-0.0652	0.466	0.961	0.133	0.02	0.387	-0.028
deepseek-v3	0.392	0.659	0.267	0.167	0.513	0.0071	0.461	0.983	1.000	0.97	—	—
gemini-2.5-pro	0.235	0.298	0.074	—	—	0.0131	2.969	—	0.000	—	—	—
gemma-3-12b	0.449	0.609	0.160	—	0.500	0.1717	0.890	0.989	1.000	6.17	—	—
gemma-3-27b	0.487	0.626	0.150	0.154	0.667	0.0000	—	0.960	1.000	1.91	0.688	-0.271
gpt-oss-120b	0.280	0.441	0.161	0.173	0.688	-0.0001	-0.000	0.991	1.000	1.09	0.657	-0.136
kimi-k2	0.484	0.675	0.191	0.019	0.887	0.0035	1.125	0.984	0.926	0.97	0.625	-0.255
llama-3.1-405b	0.392	0.464	0.102	0.062	0.500	0.0000	—	—	1.000	3.10	0.613	-0.356
llama-3.1-70b	0.414	0.501	0.212	0.049	0.551	0.0836	2.013	0.948	0.838	2.10	0.550	-0.168
llama-3.1-8b	0.388	0.532	0.186	0.107	0.807	0.0000	—	0.937	1.000	4.42	0.787	-0.139
llama-3.2-3b	0.426	0.577	0.179	—	0.577	-0.0781	2.120	0.982	1.000	3.54	—	—
llama-3.3-70b	0.499	0.767	0.267	0.100	0.797	-0.0016	-1.523	0.974	0.825	2.33	0.547	0.049
mistral-large	0.349	0.462	0.113	0.129	0.943	—	—	0.946	0.600	2.11	0.907	-0.080
mixtral-8x22b	0.460	0.593	0.136	—	0.647	-0.0252	0.363	0.984	1.000	1.60	—	—
phi-4	0.279	0.483	0.222	0.068	0.906	-0.0074	-0.000	0.960	1.000	3.30	0.721	-0.304
qwen3-next-80b	0.486	0.638	0.152	—	0.882	0.0750	0.958	0.956	1.000	1.73	—	—

arithmetic" and "I am 70% accurate on coding" cannot predict its accuracy on a task requiring both.

The Compositional Calibration Error (CCE) ranges from 0.500 (gemma-3-12b, llama-3.1-405b) to 0.943 (mistral-large), meaning mistral-large's expressed composite confidence sits 0.94 probability units above floor-level actual performance.

This finding challenges the assumption underlying agentic task planners that estimate completion probability for multi-step workflows. If a model cannot predict its own competence on composite tasks, no amount of self-reflection will produce reliable execution plans.

5.3 Finding 2: The Knowing-Doing Gap

The central finding of MIRROR is a functional dissociation between metacognitive monitoring and metacognitive control. Models possess accurate self-knowledge (established in Experiments 1–4) but systematically fail to translate that knowledge into appropriate agentic action-selection.

The Escalation Curve. We test this through four conditions that progressively externalize metacognitive control ($n = 10$ models, all paradigms):

Table 5: Escalation curve: mean CFR across four conditions ($n = 10$ models). "Adj. step" indicates significance of adjacent condition comparison.

Cond.	Description	Mean CFR	Δ from C1	Adj. step
C1	Uninformed	0.562	—	—
C2	Self-informed (MIRROR scores)	0.583	$+0.021$	ns
C3	Instructed (scores + norm. frame)	0.491	-0.071	$*$
C4	Constrained (external routing)	0.214	-0.348	$*$

Total CFR reduction C1→C4: **62%** (34.8 percentage points). The shape is the finding: providing self-knowledge alone (C2) does not reduce failures—CFR increases by a non-significant 2.1 pp, suggesting that explicit self-knowledge adds no actionable value. Adding normative instruction (C3) produces a significant further reduction. Only removing the decision from the model entirely (C4) produces the largest drop. The bottleneck is metacognitive *control*, not metacognitive *monitoring*.

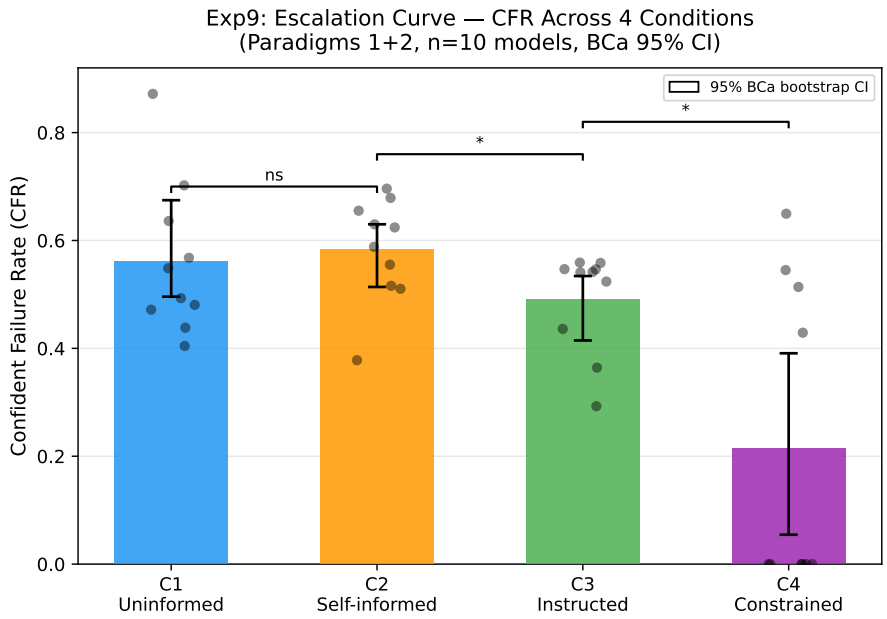

Figure 2: CFR escalation curve across four Experiment 9 conditions (mean $\pm$ 95% BCa bootstrap CI, $n = 10$ models). Self-knowledge alone (C1$\rightarrow$C2) does not reduce CFR; all 34.8 pp of improvement comes from external structure (C2$\rightarrow$C4). External constraint drives the reduction.

Universal Negative KDI. The Knowing-Doing Index (KDI = self-knowledge accuracy $-$ appropriate action rate) is negative for 9 of 10 models, ranging from -0.356 (llama-3.1-405b) to $+0.049$ (llama-3.3-70b). Table 6 shows per-model values. The lone positive outlier (llama-3.3-70b) is driven by its procedural domain, where nat_acc $= 0.059$ but the model proceeds autonomously on 70% of procedural tasks—the knowing-doing gap at its most extreme.

Null Predictive Result (Pre-registered). The MIRROR calibration gap does not independently predict agentic CFR beyond raw accuracy: Pearson $r = -0.100$, partial r (controlling for natural accuracy) $= -0.044$, $p = 0.557$. This null was pre-registered as Contingency C1. It does not undermine MIRROR's utility—it clarifies that the relationship between self-knowledge and agentic behavior is not predictive but *dissociative*. The fix is not better self-knowledge but external metacognitive scaffolding.

Deployment Implications. In a 1,000-task agentic deployment: $\sim$648 tasks fail silently under C1 (the model proceeds on weak domains despite having internal self-knowledge). Providing the model with its own MIRROR scores (C2) increases this to $\sim$583—it gets slightly *worse*. Only external constraint (C4) brings failures down to $\sim$214, a 62% reduction. Every lab deploying agentic LLMs should evaluate metacognitive *control*, not just metacognitive monitoring.

5.4 Supporting Findings

Feedback Immunity (Exp 4). AI ≈ 0 across most models—self-knowledge is not updated by experience. The two models that do adapt (deepseek-r1, llama-3.1-70b) show a domain-polarity signature: positive adaptation in factual/temporal domains, negative in procedural/social.

Adversarial Robustness (Exp 5). ARS = 0.937–0.991 across all 14 models tested. Metacognitive calibration is robust to framing attacks—models do not become more overconfident under adversarial pressure.

Table 6: Knowing-Doing Index (KDI) for all Experiment 9 models, sorted by KDI ascending. Negative KDI indicates models fail to translate self-knowledge into appropriate action-selection. CFR_{C1} = uninformed condition; CFR_{C4} = externally constrained condition.

Model	KDI	CFR_{C1}	CFR_{C4}
`llama-3.1-405b`	-0.3563	0.613	0.000
`phi-4`	-0.3041	0.721	0.650
`gemma-3-27b`	-0.2711	0.688	0.545
`kimi-k2`	-0.2553	0.625	0.514
`llama-3.1-70b`	-0.1681	0.550	0.000
`llama-3.1-8b`	-0.1389	0.787	0.000
`gpt-oss-120b`	-0.1359	0.657	0.000
`mistral-large`	-0.0803	0.907	0.000
`deepseek-r1`	-0.0278	0.387	0.000
`llama-3.3-70b`	0.0489	0.547	0.429

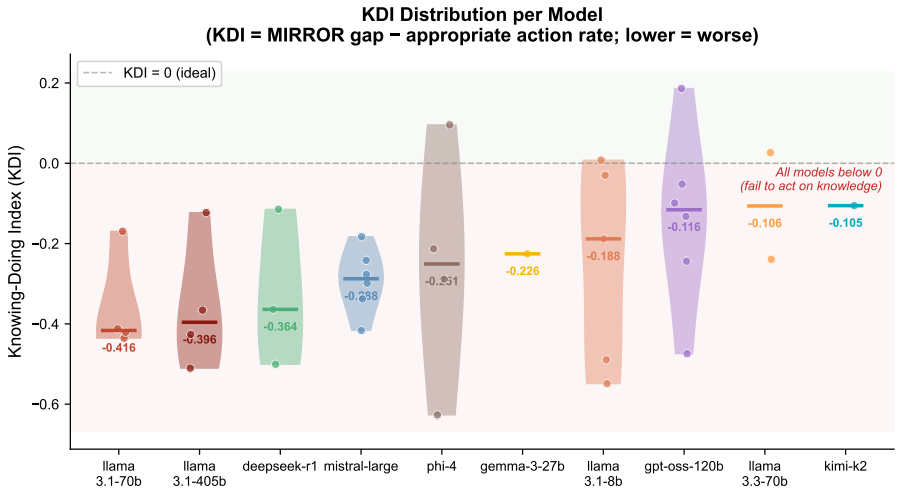

Figure 3: KDI distribution across 10 models. Nine models show uniformly negative KDI, indicating systematic failure to translate self-knowledge into appropriate action. The lone positive outlier (llama-3.3-70b, $KDI = +0.049$) reflects extreme procedural-domain miscalibration.

Flaw Detection (Exp 6b). Extreme inter-model variation: FDR ranges from 0.000 (gemini-2.5-pro—never flags flaws) to 1.000 (gemma-3-12b). deepseek-r1 achieves the lowest CFR in Experiment 9 (0.387, most cautious) but has FDR = 0.133 (worst flawed-premise detector among non-outlier models)—cautious in action but poor at epistemic judgment. Models from the same lab can differ dramatically.

Non-Monotonic Scaling (Exp 8). Within the Llama 3.1 family, 70B shows the smallest MIRROR gap; both 8B and 405B are more miscalibrated. The generation comparison (3.1-70B vs. 3.3-70B) shows capability improvement (+20.2 pp natural accuracy) without closing the calibration gap (+0.010 gap increase). Metacognitive calibration does not scale monotonically with parameters.

Weak Transfer (Exp 2). TII = 0.019–0.175 across 11 models. Self-knowledge is domain-atomic—it does not meaningfully cross domain boundaries. Verbal transfer scores (models correctly identifying their weakest domain in cross-domain tasks) are consistently low (0.039–0.140).

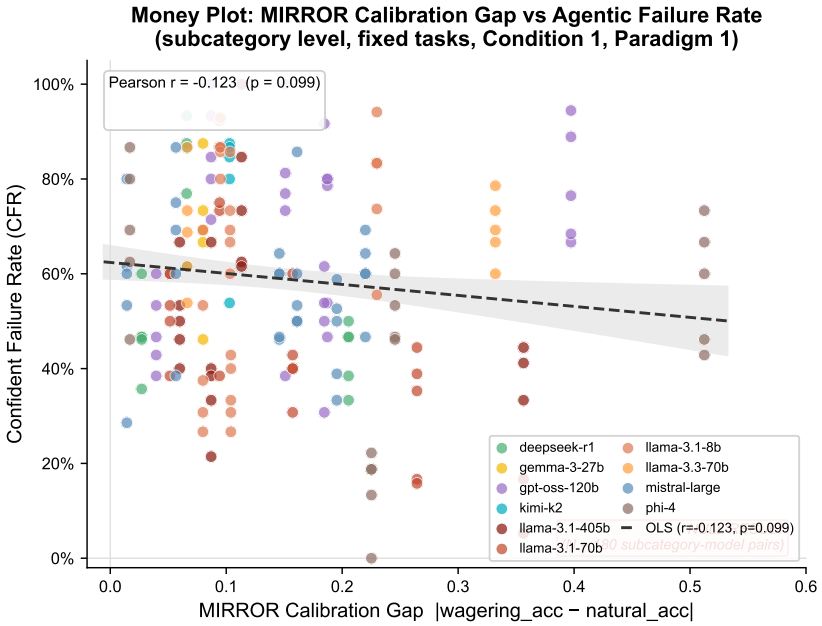

Figure 4: MIRROR gap vs. CFR at the subcategory level. The null correlation ($r = -0.100$) confirms that calibration gap does not independently predict agentic failure. Raw accuracy ($r = -0.386$, $p < 0.001$) is the dominant predictor.

6 Discussion and Limitations

6.1 Implications for Agentic System Design

The knowing-doing gap reframes the field's approach to agentic safety. Current practice focuses on improving calibration—making models more accurate in their self-assessment. MIRROR shows this is necessary but insufficient: even models with accurate domain-specific self-knowledge fail to translate that knowledge into appropriate action-selection in 56.2% of weak-domain cases (C1 baseline).

The fix is architectural, not epistemic. External routing that uses MIRROR domain scores as decision inputs reduces CFR by 62%. We recommend that every lab deploying agentic systems evaluate metacognitive *control* (not just monitoring) before deployment. MIRROR-informed routing offers a practical intervention: use per-domain calibration scores to determine which task components require external constraints, tool delegation, or human review.

6.2 The Sycophancy-Feedback Asymmetry

Experiments 4 and 6 together establish a striking asymmetry. Models do not update their calibration channels from accurate performance feedback (AI ≈ 0, Exp. 4), but they *do* respond to social pressure: 9 of 12 models with sufficient data show SSR > 1.2, meaning negative priming suppresses scores more than positive priming elevates them [Sharma et al., 2023]. The asymmetry is this: models are responsive to evaluator affect and unresponsive to data. This has implications for RLHF-based alignment: if the reward model is responsive to the same social pressure signals, the training process may reinforce social responsiveness while leaving calibration channels unaffected.

6.3 Paradigm Convergence and RLHF Confounds

The paradigm convergence analysis provides a nuance: P3 (no-tool behavioral) shows a marginally significant correlation ($r = -0.332$, $p = 0.039$) between MIRROR gap and behavioral signals, while P1 ($r = -0.114$, $p = 0.489$) and P2 ($r = -0.277$, $p = 0.088$) do not. This suggests RLHF training

may create surface-level deference cues that partially confound tool-use decisions but do not affect underlying behavioral patterns. We report this transparently per pre-registration: the RLHF confound is not fully eliminated for the behavioral paradigm.

6.4 Limitations

API-only evaluation. MIRROR requires only API access, enabling scale but preventing mechanistic analysis. Whether the knowing-doing gap reflects an information deficit or a control failure at the representation level remains open for hidden-state probing.

No human annotations. MIRROR is fully automated: questions generated and verified by LLM ensembles. This enables scale (250K instances) but prevents human–LLM comparison.

Ecological validity. Experiment 9 uses constructed agentic scenarios, not production deployments.

Model coverage. 16 models from 8 labs. Findings may not generalize to proprietary models (GPT-4o, Claude 3.5) not tested. Experiment 2 is limited to 11 models; Experiment 6c is partial for 3 models due to API unavailability.

Null predictive result. MIRROR gap does not predict CFR independently of accuracy (partial $r = -0.044$). The contribution is diagnostic (revealing the knowing-doing gap), not predictive. The 270 planned tailored tasks not completed before analysis would have increased statistical power.

MCI design artifact. The pre-registered MCI requires strong-strong and weak-weak composite trials. Task generation rarely produced both simultaneously, rendering MCI = 0 partially a design artifact. CCE is the recommended primary Level 2 metric.

6.5 Future Work

Mechanistic probing: hidden-state analysis on open-weight models (Llama 3.1, Phi-4, Gemma-3) to determine if the knowing-doing gap reflects information deficit or control failure at the representation level. *Human baseline*: compressed MIRROR evaluation on human participants for cross-species metacognitive comparison. *Behavioral fine-tuning*: fine-tune on MIRROR behavioral data and test transfer to agentic contexts. *Multi-agent cascades*: single-point metacognitive failures compounding over multi-step chains. MIRROR-*informed routing*: production deployment of domain-score-based routing to validate the C4 intervention in real-world settings.

7 Conclusion

We have introduced MIRROR, a hierarchical benchmark for metacognitive calibration in LLMs, and reported results across eight experiments spanning 16 models from 8 labs. The central finding is precise: models possess domain-specific self-knowledge that accurately identifies weak domains, but this self-knowledge does not transfer across domains, does not compose correctly, does not update from feedback, and does not translate into appropriate agentic action. The Knowing-Doing Index is negative for 9 of 10 models. Scale does not resolve this.

What works is external structure: architectural constraint reduces the Confident Failure Rate by 62%, from 0.562 to 0.214. The implication for AI safety is that trust in an LLM's self-reported uncertainty—a common design assumption in agentic systems—is insufficient. Calibration must be measured externally, then enforced architecturally. MIRROR provides the measurement instrument.

Acknowledgments

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A Scaling Analysis

Figure 5 shows the Experiment 8 scaling analysis. Within the Llama 3.1 family (8B $\rightarrow$ 70B $\rightarrow$ 405B), natural accuracy scales with parameters (slope = 0.028/log-unit, $R^2 = 0.823$) but the MIRROR gap does not (slope ≈ 0, $R^2 = 0.000$). The generation comparison (3.1-70B vs. 3.3-70B) shows capability improvement (+20.2 pp natural accuracy, +21.2 pp wagering accuracy) without closing the calibration gap (+0.010 MIRROR gap increase).

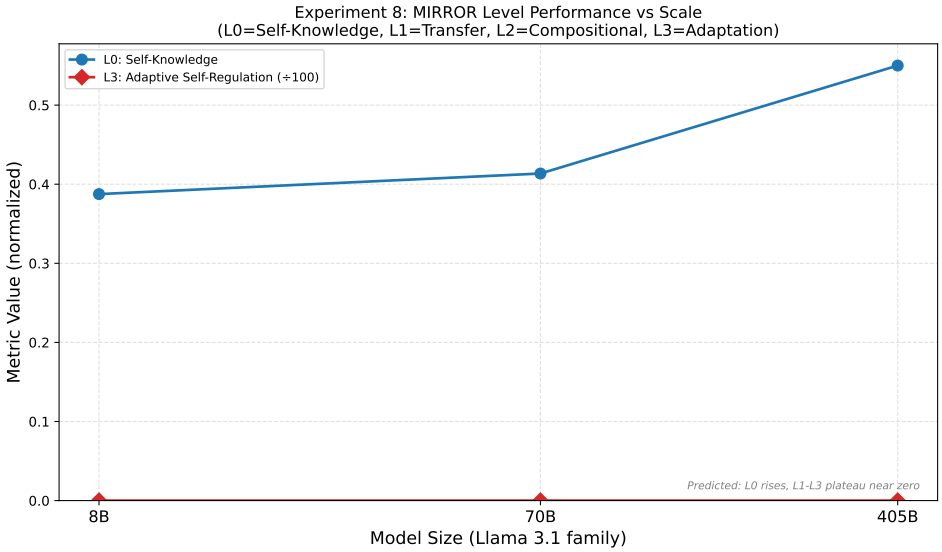

Figure 5: Scaling analysis: MIRROR metrics across model size (Llama 3.1 family) and generation (3.1 vs. 3.3 at 70B). Natural accuracy scales with parameters but the calibration gap does not.

B Per-Paradigm Escalation

Figure 6 shows the CFR escalation curve broken down by paradigm. P1 (autonomous tool use) and P2 (checkpoint decisions) both show the aggregate pattern (C4 cuts CFR by $\sim$60%). P3 (behavioral, no tools) has no C4 condition. The aggregate 62% reduction is driven by both tool-enabled paradigms.

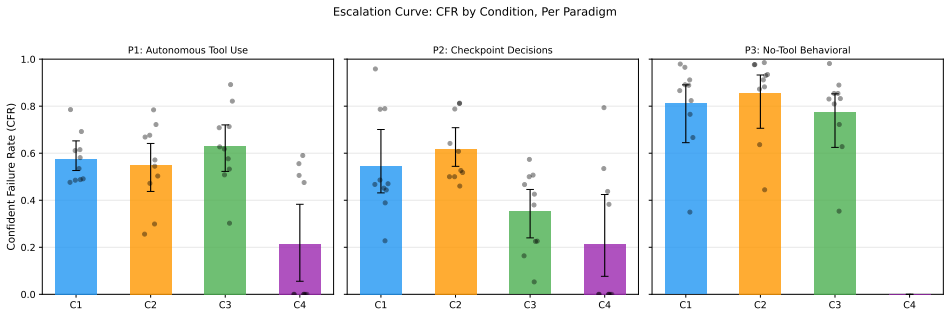

Figure 6: Per-paradigm CFR escalation curves. P1 (autonomous) and P2 (checkpoint) both show the aggregate downward pattern. P3 (no tools) has no C4 condition.